

Mrinank Gaur

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(Holding Indian Passport valid upto 26/07/2033)

PROFESSIONAL SUMMARY

Software Developer with hands-on experience in Java, Python, JavaScript, SQL, and full-stack development. Strong in DSA, problem-solving, Linux, and backend development, with experience building practical projects using React, Node.js, and FastAPI. Quick learner focused on writing clean, efficient, and reliable code.

PROJECTS AND EXPERIENCE

Digital Design Intern May 2024 July 2024

Maven Silicon Chennai, India

- Designed and implemented an **APB-AHB bridge** for System-on-Chip (SoC) integration.
- Developed **RTL designs in Verilog** and verified bus protocols.
- Performed simulation, synthesis, and functional verification using **Cadence tools**.

Real-Time Mood Detection via Gait Analysis

Python, FastAPI, Next.js, WebSockets, ESP32 | [GitHub](#) | Nov 2025 – April 2026

- Developed a **real-time biometric mood** detection system using **ESP32 force/IMU sensors** and a **Python FastAPI server**.
- Implemented a **machine learning processing pipeline** for **gait analysis** using **Scikit-Learn (Random Forest)**, transmitting continuous predictions with low latency via **WebSockets**.
- Built **responsive Next.js web interface** featuring live dynamic pressure **heatmaps**, **foot orientation metrics**, and a **real-time streaming** dashboard.
- Achieved **>90% accuracy** in Mood Detection.

Shopalytics B2B SaaS for Shopify Stores

Next.js, Node.js, Prisma, PostgreSQL | [GitHub](#) | [Shopalytics](#) | Aug 2025 – Sept 2025

- Developed **B2B SaaS** integrating Shopify APIs for automated store data ingestion and synchronization.
- Engineered multi-tenant backend with **Prisma + PostgreSQL** and secure role-based authentication.
- Delivered business insights via **interactive dashboards** for revenue, customers, and order trends.

Flex-Sensor Smart Glove for Sign Language Recognition & Speech Translation

ESP32, Python, LSTM, Next.js, MPU6050 | [Github](#) | Jul 2025 – Nov 2025

- Built a wireless smart glove with **5 flex sensors + IMU**, achieving **95% stable sensor calibration** and **50–100 Hz** real-time data streaming via ESP32.
- Trained an **LSTM gesture model** reaching **80.4% test accuracy** across 30+ ASL gestures, with **<100 ms** end-to-end inference latency.
- Developed a **Next.js web app** delivering **100% real-time gesture visualization**, **5-language translation**, and **instant speech output** for seamless sign-to-speech communication.

Boardly Collaborative Whiteboard

Next.js, TypeScript, LiveBlocks, Convex | [GitHub](#) | [Boardly](#) | May 2025 - Jul 2025

- Built **real-time collaborative whiteboard** with live cursors and **multi-user** editing via **LiveBlocks (websocket service)**.
- Developed drawing toolkit with **5+ tools (shapes, text, notes, freehand)** and layer management.
- Implemented **authentication, board sharing, and responsive UI** with modern design patterns.

Automatic Number Plate Recognition

Python, OpenCV, EasyOCR, ESP32 | [GitHub](#) | Jan 2025 - Mar 2025

- Developed real-time license plate detection system **using ESP32-CAM** and Python server with **95% accuracy**.
- Implemented OpenCV image processing pipeline with **EasyOCR**, reducing recognition **latency under 2s**.
- Built responsive web interface for image capture, processing, and real-time preview across formats.

TECHNICAL SKILLS

- Languages:
 - C/C++
 - Python
 - Java
 - SQL
 - HTML/CSS
 - TypeScript
- Cloud:
 - AWS
- Frameworks:
 - React
 - Node.js
 - Express.js
 - Material-UI
 - Gen AI
- Developer Tools:
 - Linux
 - Git
 - VSCode
 - Claude Code
- Libraries:
 - OpenCV
 - Pandas
 - NumPy
 - Matplotlib
- Other:
 - MongoDB
 - Convex
 - Clerk
 - LiveBlocks
 - JWT
 - Nodemailer

EDUCATION

- Vellore Institute of Technology 2022–2026
- B.Tech in Electronics and Communication Engineering CGPA: **8.94/10**
- The Khaitan School 2022
- Class XII (CBSE) Percentage: **89.2%**
- Class X (CBSE) Percentage: **96.2%**

CLUBS / EXTRACURRICULAR

- Robotics Engineer Technocrats Robotics, VIT University, Oct 2023 May 2025, Vellore, India
- Designed and calibrated 8-axis movement control for autonomous rover.
- Collaborated on modular software for motion planning and diagnostics.